

Mayer Path Homology

CMSE TDA SEMINAR

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Outline



Part 1

Mayer Homology

Mayer Homology

An N -chain complex consists of a graded $\mathbb{K}$ -linear space $C = (C_n)_{n \geq 0}$, equipped with a N -nilpotent linear map $d : C_n \rightarrow C_{n-1}$ satisfying $d^N = 0$. The linear map $d : C_n \rightarrow C_{n-1}$ is called the N -differential or N -boundary operator.

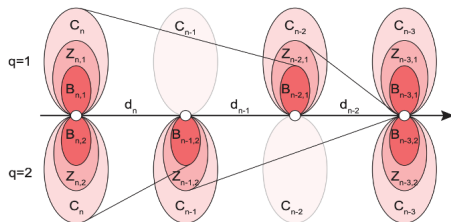
For a N -chain complex (C_*, d) and $1 \leq q \leq N - 1$, the space of q -th n -cycles defined as

$$Z_n^{N,q} = \{x \in C_n \mid d^q x = 0\}$$

and the space of the q -th n -boundaries is defined as

$$B_n^{N,q} = \{d^{N-q} x \mid x \in C_{n+N-q}\}$$

Mayer Homology



The Mayer homology of the N -chain complex (C_*, d) is defined as

$$H_n^{N,q}(C_*, d) := Z_n^{N,q} / B_n^{N,q}, \quad n \geq 0$$

The Mayer Betti Numbers are the dimension of the mayer homology groups at corresponding dimensions

$$\beta_n^{N,q} = \dim(H_n^{N,q}(C_*, d))$$

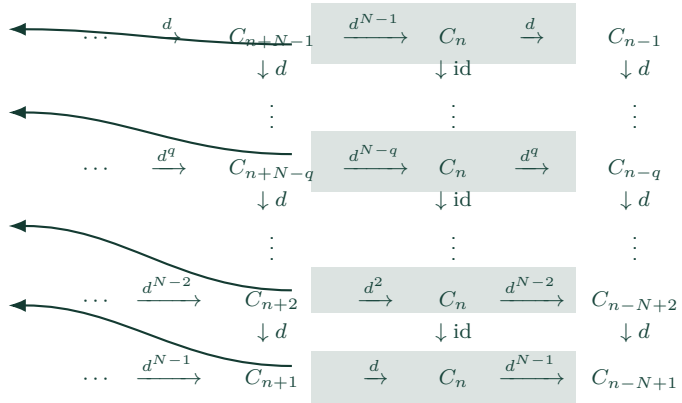
Mayer homology for general N

$$\begin{aligned} H^{N,1} &= Z^{N,1}/B^{N,1} \\ &= \ker d / \operatorname{im} d^{N-1} \end{aligned}$$

$$\begin{aligned} H^{N,q} &= Z^{N,q}/B^{N,q} \\ &= \ker d^q / \operatorname{im} d^{N-q} \end{aligned}$$

$$\begin{aligned} H^{N,N-2} &= Z^{N,N-2}/B^{N,N-2} \\ &= \ker d^{N-2} / \operatorname{im} d^2 \end{aligned}$$

$$\begin{aligned} H^{N,N-1} &= Z^{N,N-1}/B^{N,N-1} \\ &= \ker d^{N-1} / \operatorname{im} d \end{aligned}$$



History of Mayer Homology

- ▶ *Mayer*: A new homology theory, 1942
 - ▶ Introduced the notion of homology of $d^N = 0$
- ▶ *Spainer*: The Mayer Homology Theory, 1949
 - ▶ If coefficient field has characteristic prime p , Mayer homology will coincide with standard homology
- ▶ *Dunois-Violette*: $d^N = 0$, 1990
 - ▶ Formalized Mayer's construction in the language of N -complexes and generalized homology
- ▶ *Kapranov*: On the q th analog of homological algebra, 1996
 - ▶ Construction of simplicial complex by using N -th primitive root of unity
- ▶ *Tikaradze*: Homological construction on N -complexes, 2001
 - ▶ Euler Poincare formula that is generalized Euler char for Mayer groups

Mayer Homology

$\partial_i = ([v_0, \dots, v_n]) = [v_0, \dots, v_{i-1}, v_{i+1}, \dots, v_n]$ is the face map

Standard vs Mayer Homology

► *Standard Chain Map :*

$$d_n = \sum_i (-1)^i \partial_i \quad d^2 = 0$$

► *Mayer Chain Map:*

$$d_n = \sum_i (\xi)^i \partial_i \quad d^N = 0$$

$$d^N = \left(\prod_{k=1}^N (1 + \xi + \dots + \xi^{k-1}) \right) \sum_{j_1 < \dots < j_N} \xi^{j_1 + \dots + j_N - \frac{N(N-1)}{2}} \partial_{j_1} \dots \partial_{j_N}$$

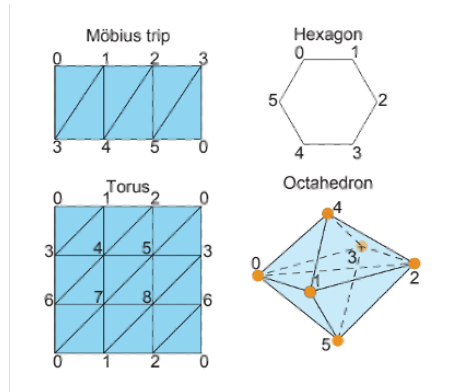
Mayer Betti Number

- Mayer Betti Numbers are the dimension of the homology groups on the corresponding dimensions

$$\beta_n^{N,q} = \dim H_n^{N,q}$$

simplicial complexes	$\beta_{0,1}$	$\beta_{1,1}$	$\beta_{2,1}$	$\beta_{0,2}$	$\beta_{1,2}$	$\beta_{2,2}$
$\Delta[3]$	1	1	0	0	2	0
$\partial\Delta[3]$	1	2	0	0	2	1
Hexagon	6	0	0	0	6	0
Möbius trip	1	6	0	0	6	1
Torus	1	18	0	0	9	10
Octahedron	1	3	1	0	2	3

Table of Betti numbers for $N = 3$



Triangulation used to compute Mayer betti numbers

Mayer Betti Number

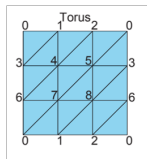


Figure: Triangulation of Torus T_2^1

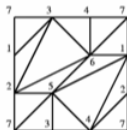


Figure: Minimal Triangulation of torus T_2^2

Mayer Betti Numbers

$$\triangleright \beta_0^{3,1}(T_2^1) = 1 \quad \beta_1^{3,1}(T_2^1) = 18 \quad \beta_2^{3,1}(T_2^1) = 0$$

$$\triangleright \beta_0^{3,2}(T_2^1) = 0 \quad \beta_1^{3,2}(T_2^1) = 9 \quad \beta_2^{3,2}(T_2^1) = 10$$

$$\triangleright \beta_0^{3,1}(T_2^2) = 0 \quad \beta_1^{3,1}(T_2^2) = 14 \quad \beta_2^{3,1}(T_2^2) = 0$$

$$\triangleright \beta_0^{3,2}(T_2^2) = 0 \quad \beta_1^{3,2}(T_2^2) = 7 \quad \beta_2^{3,2}(T_2^2) = 7$$

Persistent Mayer Homology

- Mayer homology $H_*^{N,q}$ is a functor from $N\text{chain}$ to Vec for each $1 \leq q \leq N - 1$

$$\text{Filtration: } C_*(K_{a_1}) \subset \cdots \subset C_*(K_{a_m})$$

$$\text{Induced Maps: } H_*^{N,q}(C_*(K_{a_1})) \subset \cdots \subset H_*^{N,q}(C_*(K_{a_m}))$$

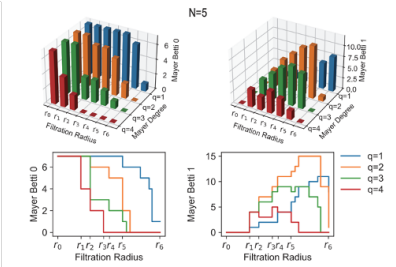
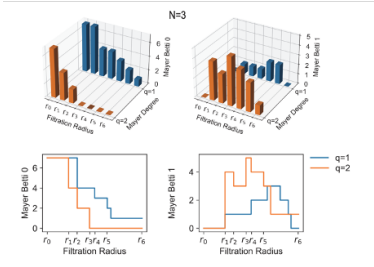
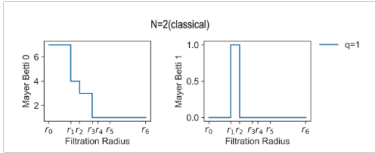
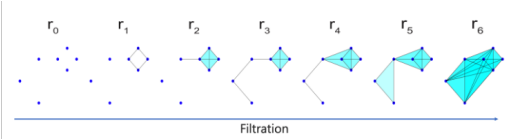
- Persistent classes (a, b) -persistent Mayer homology

$$H_{n,(a,b)}^{N,q} := \text{im}\{H_n^{N,q}(C_*(K_a)) \rightarrow H_n^{N,q}(C_*(K_b))\}$$

- The rank of $H_{n,(a,b)}^{N,q}$ is the (a, b) -persistent Betti numbers.

Li Shen, Jian Liu, Guo-Wei Wei - Persistent Mayer homology and persistent Mayer Laplacian, 2023

Persistent Mayer Homology



Li Shen, Jian Liu, Guo-Wei Wei - Persistent Mayer homology and persistent Mayer Laplacian, 2023

Part 2

Path Complex

Path Homology

Why Path Homology?

Motivation

Classical notions of homology for graphs often fail to capture higher-dimensional structure.

Limitations of Existing Approaches

- ▶ **Graph homology:**
 - ▶ Graph as 1D simplicial complex
 - ▶ Trivial for $n \geq 2$
- ▶ **Clique (simplicial) homology:**
 - ▶ Uses complete subgraphs of graphs
- ▶ **Hochschild homology:**
 - ▶ Trivial for $n \geq 2$ on acyclic digraphs

Key Advantage

Path complexes can produce **nontrivial homology groups** in higher degrees.

⇒ Captures richer structure in directed graphs

Alexander Grigor'yan, Yong Lin, Yuri Muranov, Shing-Tung Yau, Homologies of path complexes and digraphs, 2012

Paths on Finite Sets

Elementary p -paths

Let V be a nonempty finite set of vertices. For $p \in \mathbb{Z}_{\geq 0}$, an *elementary p -path* $e_{i_0 \dots i_p}$ on V is any sequence $i_0 \dots i_p$ of $p+1$ vertices in V .

When $p = -1$, the set of elementary p -paths is empty.

Linear space generated by elementary paths

The vector space over $\mathbb{K}$ generated by all elementary p -paths on V is denoted

$$\Lambda_p(V) = \Lambda_p(V, \mathbb{K}).$$

Any $v \in \Lambda_p(V)$ can be written as

$$v = \sum_{i_0, \dots, i_p \in V} c^{i_0 \dots i_p} e_{i_0 \dots i_p}, \quad c^{i_0 \dots i_p} \in \mathbb{K}.$$

Boundary map

The boundary map

$$\partial_p : \Lambda_p(V) \rightarrow \Lambda_{p-1}(V)$$

is defined on elementary paths by

$$\partial_p(e_{i_0 \dots i_p}) = \sum_{j=0}^p (-1)^j e_{i_0 \dots \hat{i}_j \dots i_p},$$

where $\hat{i}_j$ means that the vertex i_j is omitted.

Idea

This is the analogue of the usual simplicial boundary, but defined for arbitrary vertex sequences.

Regular and Irregular Paths

Regular and Irregular paths

An elementary path is regular if any consecutive terms are different.

$$\mathcal{R}_p = \text{span} \langle \text{regular} \rangle \quad \mathcal{N}_p = \text{span} \langle \text{non-regular} \rangle \quad \Lambda_p = \mathcal{R}_p \oplus \mathcal{N}_p$$

$$\partial \mathcal{R}_p \not\subset \mathcal{R}_{p-1} : \quad \partial(e_{121}) = e_{21} - e_{11} + e_{12},$$

Regular boundary map

It is shown by Grigor'yan et al. that the boundary map on $\Lambda_p/\mathcal{N}_p$ is well defined. The induced map

$$\bar{\partial} : \mathcal{R}_p \rightarrow \mathcal{R}_{p-1}$$

is called the *regular boundary map*. Hence the chain complex of V can be written as

$$(\mathcal{R}_p, \bar{\partial})_p.$$

From now on, we denote the regular boundary map simply by ∂ .

Path Complexes

Definition

A *path complex* over a set V is a nonempty collection P of elementary paths on V such that for every $n \in \mathbb{N}$,

$$i_0 \cdots i_n \in P \implies i_0 \cdots i_{n-1} \in P \text{ and } i_1 \cdots i_n \in P.$$

Allowed and non-allowed paths

The elementary paths belonging to P are called *allowed paths*. Elementary paths on V that do not belong to P are called *non-allowed paths*.

Examples of Path Complexes

From simplicial complexes

Let K be an abstract finite simplicial complex on a finite vertex set V . Since K is closed under taking subsets, after fixing an ordering of vertices, each simplex can be represented as an elementary path.

Denote the resulting path complex by $P(K)$. Then the allowed n -paths in $P(K)$ are precisely the n -simplices of K .

From digraphs

A path complex can also be derived from a digraph G by taking all directed paths in G as allowed paths.

Allowed-path spaces

For a path complex P , define the $\mathbb{K}$ -linear space

$$\mathcal{A}_n = \mathcal{A}_n(P) = \text{span}\{e_{i_0 \dots i_n} \mid i_0 \dots i_n \in P\}.$$

Path Homology

Problem

In general, $(\mathcal{A}_n, \partial)$ does not form a chain complex.

∂ -invariant paths

$$\Omega_n = \Omega_n(P) = \{v \in \mathcal{A}_n \mid \partial v \in \mathcal{A}_{n-1}\}.$$

These are called the ∂ -invariant n -paths.

Path homology

$$\partial\Omega_n \subset \Omega_{n-1},$$

(Ω, ∂) is a chain complex. The homology of (Ω, ∂) is called the *path homology* of the path complex P .

Path Complexes

Motivation

We first need to identify its elementary objects. In simplicial homology, these are simplices:

- ▶ vertices, edges, triangles, tetrahedra, ...

Grigor'yan et al, 2012

Elements of Ω_2 of path complex $P(G)$ induced by finite digraph G is a linear combination of double edge, triangles and square.

Basic edge configurations

- ▶ $e_{ij} + e_{ji}$: a double edge,
- ▶ $e_{ij} + e_{jk} + e_{ik}$: a triangle,
- ▶ $e_{ij} + e_{jk} + e_{il} + e_{lk}$: a square.

Path Complexes

Grigor'yan, Advances in path homology theory of digraphs, 2022

If G contains neither multisquare nor double arrow, then there is a basis in $\Omega_3(G)$ that consists of trapezohedral paths T_m with $m \geq 2$ and their merging images.

Trapezohedral

For any integer $m \geq 2$, define a trapezohedron T_m of order m as follows: T_m is a digraph of $2m + 2$ vertices

$$a, b, i_0, \dots, i_{m-1}, j_0, j_1, \dots, j_{m-1}$$

and $4m$ arrows

$$a \rightarrow i_k \rightarrow j_k \rightarrow b, \quad i_k \rightarrow j_{k+1}$$

for all $k = 0, \dots, m-1 \pmod{m}$

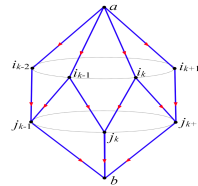
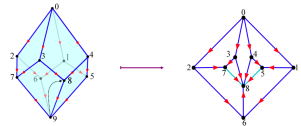
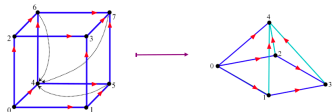
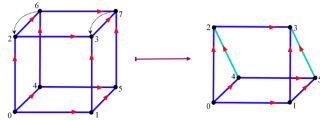
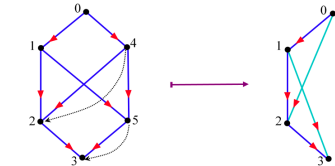


Figure: Trapezohedron

Merging Images of Trapezohedron

Li and Shen - ∂ -invariant path generators for digraphs, 2026

The assumption for the basis is eliminated.



Part 3

Mayer Path Homology

Mayer Path Homology

Definition of Mayer boundary

Define the boundary operator $\partial : \Lambda_p \rightarrow \Lambda_{p-1}$ by

$$\partial(e_{i_0 i_1 \dots i_p}) = \sum_{j=0}^p \xi^j e_{i_0 \dots \hat{i}_j \dots i_p},$$

where ξ is an N -th root of unity and $e_{i_0 \dots \hat{i}_j \dots i_p}$ denotes the path obtained by omitting the j -th index.

$(\mathcal{A}_n, \partial)$ is not N -chain complex. For $P = \{e_{12}, e_{23}, e_{123}\}$

$$\partial(e_{123}) = e_{23} + \xi e_{13} + \xi^2 e_{12},$$

∂ -invariant allowed paths

For $N \geq 2$ and $1 \leq q \leq N-1$, define the space of ∂ -invariant n -paths of level N and step q as

$$\Omega_n^{N,q} = \Omega_n^{N,q}(P) = \{v \in \mathcal{A}_n \mid \partial^q v \in \mathcal{A}_{n-q}\}.$$

The intersection is called ∂ -invariant n -paths and denoted as follows:

$$\Omega_n^N = \bigcap_{1 \leq q \leq N-1} \Omega_n^{N,q}(P) = \{v \in \mathcal{A}_n \mid \partial^q v \in \mathcal{A}_{n-q}\}.$$

For $q = N-1$, we have $\partial^{N-1}(\partial v) = 0 \in \mathcal{A}_{n-N}$. Thus, $\partial(v) \in \Omega_{n-1}^{N,q}$ for all $1 \leq q \leq N-1$.

Mayer Path Homology

Definition of Mayer Path Homology

Let P be a path complex associated with the N -chain complex (Ω_n^N, ∂) . For any $1 \leq q \leq N - 1$, define the following spaces:

- The space of q -th n -cycles:

$$Z_n^{N,q} = \{x \in \Omega_n^N \mid \partial^q(x) = 0\}.$$

- The space of q -th n -boundaries:

$$B_n^{N,q} = \{\partial^{N-q}(x) \mid x \in \Omega_{n+N-q}\}.$$

Then, the **Mayer Path Homology** is defined as

$$H_n^{N,q}(P) := Z_n^{N,q} / B_n^{N,q} = \ker(\partial_n^q) / \operatorname{im}(\partial_{n+N-q}^{N-q}).$$

The rank of $H_n^{N,q}(P)$ is called the **Mayer Path Betti number**.

A Comparison Example: T_1 and T_2

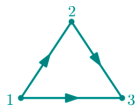


Figure: T_1

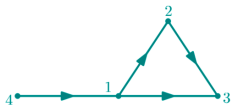


Figure: T_2

Standard Homology

For T_1 , $\Omega_0^2 = \mathcal{A}_0 = V_1$, $\Omega_1^2 = \mathcal{A}_1 = E_1$ and $\Omega_2^2 = \mathcal{A}_2 = \{e_{1,2,3}\}$

$$Z_0^{2,1}(T_1) = \langle V_1 \rangle, \quad B_0^{2,1}(T_1) = \langle e_2 - e_1, e_3 - e_1 \rangle$$

$H_0^{2,1}(T_1) \simeq \mathbb{C}$. $H_n^{2,1}(T_1) = 0$ for $n \geq 1$

For T_2 , $\Omega_0^2 = \mathcal{A}_0 = V_2$, $\Omega_1^2 = \mathcal{A}_1 = E_2$ and $\Omega_2^2 = \{e_{1,2,3}\}$

$$Z_0^{2,1}(T_2) = \langle \gamma_2 \rangle, \quad B_0^{2,1}(T_2) = \langle e_2 - e_1, e_3 - e_1, e_4 - e_1 \rangle$$

$H_0^{2,1}(T_2) \simeq \mathbb{C}$. $H_n^{2,1}(T_2) = 0$ for $n \geq 1$

A Comparison Example: T_1 and T_2

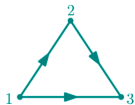


Figure: T_1

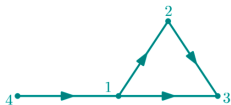


Figure: T_2

Mayer Homology $N = 3, q = 1$

For T_1 , $\Omega_0^3 = \mathcal{A}_0 = V_1$, $\Omega_1^3 = \mathcal{A}_1 = E_1$ and $\Omega_2^3 = \mathcal{A}_2 = \{e_{1,2,3}\}$

$$Z_0^{3,1}(T_1) = \langle \gamma_1 \rangle, \quad B_0^{3,1}(T_1) = \langle -\xi e_1 - e_2 - \xi^2 e_3 \rangle$$

$$Z_1^{3,2}(T_1) = \langle e_{1,2} - e_{1,3} + e_{2,3} \rangle, \quad B_1^{3,2}(T_1) = \langle \xi^2 e_{1,2} + \xi e_{1,3} + e_{2,3} \rangle$$

$$H_0^{3,1}(T_1) \simeq \mathbb{C}^2 \text{ and } H_1^{3,1}(T_1) \simeq 0$$

For T_2 , $\Omega_0^3 = \mathcal{A}_0 = V_2$, $\Omega_1^3 = \mathcal{A}_1 = E_2$ and $\Omega_2^3 = \{e_{1,2,3}\}$

$$Z_0^{3,1}(T_2) = \langle \gamma_2 \rangle, \quad B_0^{3,1}(T_2) = \langle -\xi e_1 - e_2 - \xi^2 e_3 \rangle, \quad Z_1^{3,1}(T_2) = 0$$

$$, H_0^{3,1}(T_2) \simeq \mathbb{C}^3, \text{ and } H_1^{3,1}(T_2) \simeq 0$$

Structure of $\Omega_n^{N,1}$

Karaguler, Mayer Path Homology, 2026

Elements of $\Omega_2^N = \Omega_2^{N,1}$ of path complex $P(G)$ induced by finite digraph G can be represented as a linear combination of double edge, triangle and square for $N \geq 2$.

Next question

Does $\Omega_3^{N,1}$ have the same structure with Ω_3^2 that is formed by trapezohedrons and their merging images?

Structure of $\Omega_n^{N,1}$

Karaguler, Mayer Path Homology, 2026

Elements of $\Omega_2^N = \Omega_2^{N,1}$ of path complex $P(G)$ induced by finite digraph G can be represented as a linear combination of double edge, triangle and square for $N \geq 2$.

Next question

Does $\Omega_3^{N,1}$ have the same structure with Ω_3^2 that is formed by trapezohedrons and their merging images?

Answer

ALMOST YES

Cluster Structure and Basis Construction

Cluster Paths

A p -path

$$v = \sum v^{i_0, \dots, i_p} e_{i_0, \dots, i_p}$$

is called an (a, b) -cluster if all elementary paths with nonzero coefficients satisfy $i_0 = a, i_p = b$.

A path is called a *cluster* if it is an (a, b) -cluster for some $a, b \in V(G)$.

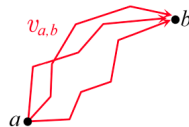


Figure: (a, b) -cluster

Cluster-Graph Construction

Associate to an (a, b) -cluster with a graph Γ :

- Vertices of Γ correspond to elementary paths $e_{a,i,j,b}$,
- Edges are assigned according to relations between these paths,
- The edge set is partitioned into colored classes E_1, E_2 .

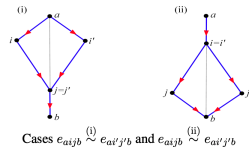


Figure: E_1 and E_2 labeled edges

Example of Construction of Generators

Cluster Structure and Basis Construction

$$\begin{aligned}\gamma_1 &= (S, T, N_w, T), & \gamma_2 &= (S, S, N_w, T), & \gamma_3 &= (S, W, N_w, T), \\ \gamma_4 &= (T, N_w, T, S), & \gamma_5 &= (T, N_w, S, S), & \gamma_6 &= (T, N_w, W, S), \\ \gamma_7 &= (S, N_w, N_w, S) & \gamma_8 &= (T, S, S, T), & \gamma_9 &= (T, T, T, T)\end{aligned}$$

Cluster Structure and Basis Construction

$$\begin{aligned}\gamma_1 &= (S, T, N_w, T), & \gamma_2 &= (S, S, N_w, T), & \gamma_3 &= (S, W, N_w, T), \\ \gamma_4 &= (T, N_w, T, S), & \gamma_5 &= (T, N_w, S, S), & \gamma_6 &= (T, N_w, W, S), \\ \gamma_7 &= (S, N_w, N_w, S) & \gamma_8 &= (T, S, S, T), & \gamma_9 &= (T, T, T, T)\end{aligned}$$

Coexistence Condition

The existence of e_{ab} and number of e_{a*b} determines the following groups

$$T_1 = (\gamma_1, \gamma_4, \gamma_7) \quad T_2 = (\gamma_2, \gamma_5, \gamma_7) \quad T_3 = (\gamma_3, \gamma_6, \gamma_7) \quad T_4 = (\gamma_7) \quad T_5 = (\gamma_9) \quad T_6 = (\gamma_8)$$

Basis Construction

Karaguler, Mayer Path Homology, 2026

For $N = 2$ and $i = 1, 2, 3$:

$$\gamma_i - (\text{even number of intermediate } \gamma_7\text{-vertices}) - \gamma_i,$$

$$\gamma_{i+3} - (\text{even number of intermediate } \gamma_7\text{-vertices}) - \gamma_{i+3}.$$

For $i = 1, 2$

$$\gamma_1 - (\text{odd number of intermediate } \gamma_7\text{-vertices}) - \gamma_4$$

$$(4m + 2)\gamma_7 - \gamma_8 - \gamma_9$$

Basis Construction

Karaguler, Mayer Path Homology, 2026

For $N \geq 3$ and $i = 1, 2, 3$:

$$\gamma_i - (\text{even number of intermediate } \gamma_7\text{-vertices}) - \gamma_i,$$

$$\gamma_{i+3} - (\text{even number of intermediate } \gamma_7\text{-vertices}) - \gamma_{i+3}.$$

$$\gamma_1 - (\text{odd number of intermediate } \gamma_7\text{-vertices}) - \gamma_4$$

$$(4m + 2)\gamma_7 - \gamma_9$$

Basis Construction

Karaguler, Mayer Path Homology, 2026

For $N \geq 3$ and $i = 1, 2, 3$:

$$\gamma_i - (\text{even number of intermediate } \gamma_7\text{-vertices}) - \gamma_i,$$

$$\gamma_{i+3} - (\text{even number of intermediate } \gamma_7\text{-vertices}) - \gamma_{i+3}.$$

$$\gamma_1 - (\text{odd number of intermediate } \gamma_7\text{-vertices}) - \gamma_4$$

$$(4m + 2)\gamma_7 - \gamma_9$$

Karaguler, Mayer Path Homology

The basis of Ω_3^N is the basis of the $\Omega_3^{N,1}$

H_1^N of Path Complexes

$$\cdots \rightarrow \Omega_3^N \rightarrow \Omega_2^N \rightarrow \Omega_1^N \rightarrow \Omega_0^N \rightarrow 0$$

$$H_1^{N,q} = Z_1^{N,q} / B_1^{N,q} = \ker \partial^q / \operatorname{imd}^{N-q}$$

$$q = 1$$

$$H_1^{N,1} = \ker \partial / \operatorname{imd}^{N-1}$$

$$q = 2$$

$$\begin{aligned} H_1^{N,2} &= \ker \partial^2 / \operatorname{imd}^{N-2} \\ &= \Omega_1^N / \operatorname{im}(\Omega_{N-1}^N) \end{aligned}$$

Observation

When $N = 3$, we need to know Ω_2^3 and $Z_1^{3,1}$.

Basis of the 1-Cycle Space

Consider the 1-cycle space for $\partial_1 : \Omega_1^2 \rightarrow \Omega_0^2$

$$Z_1 \subseteq \Omega_1, \quad Z_1 = \ker(\partial_1),$$

Spanning Tree Construction

Let G_u be the underlying undirected graph of a digraph G , and let T be a rooted spanning tree of G_u with root r . Define $\overline{T} := G_u \setminus T$.

For each edge $e = (v_1, v_2) \in \overline{T}$, define a cycle c_e by combining:

- the edge e , the path from v_1 to r , and the path from v_2 to r .

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Cycle Basis

Each edge $e \in \overline{T}$ determines a unique undirected cycle in G_u .

The collection

$$\{c_e \mid e \in \overline{T}\}$$

forms a basis of the 1-cycle space of G_u . Each cycle c_e induces a cycle in $\Omega_1(G)$ by assigning coefficients 1 or -1 according to the orientation of edges in G .

Tamal et al - An efficient algorithm for 1-dimensional (persistent) path homology, 2022

The cycles $\{c_e \mid e \in \overline{T}\}$ form a basis of $Z_1 = \ker(\partial_1)$ over any coefficient field $\mathbb{K}$.

An example of 1-cycle is not in $Z_1^{N,1}$

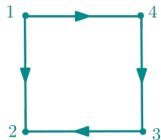


Figure: The digraph L

$Z_1^{N,1}$ for $N = 2, 3$

Spanning tree of L is $\{e_{1,4}, e_{4,3}, e_{3,2}\}$. There is only one 1-cycle
 $c_{e_{1,2}} = -e_{1,2} + e_{1,4} + e_{4,3} + e_{3,2}$

$$\partial(c_{e_{1,2}}) = (1 + \xi)(e_3 + e_4)$$

For $N = 2$, $\partial(c_{e_{1,2}}) = 0$. For $N = 3$, $\partial(c_{e_{1,2}}) \neq 0$

Kernel of ∂_1 for $N \geq 3$

Karaguler, Mayer Path Homology, 2026

Assign to each edge a weight $x_j \in \{1, \xi\}$ depending on orientation, where ξ is a primitive N th root of unity. Define coefficients recursively:

$$y_j = -\frac{x_{j-1}^*}{x_j} y_{j-1}.$$

Then a weighted cycle $\sum_{j=1}^n y_j v_j$ lies in $\ker(\partial_1)$ if and only if $(-1)^n \prod_{j=1}^n \frac{x_j^*}{x_j} = 1$.

Admissible cycles

Let u_1, u_2 count occurrences of $\frac{x_j^*}{x_j} = \xi$ and ξ^{-1} .

Then the kernel condition is:

- n even: $u_1 - u_2 \equiv 0 \pmod{N}$
- n odd: N even and $u_1 - u_2 \equiv \frac{N}{2} \pmod{N}$

Cycles satisfying this are called **admissible cycles**.

Structure of $\ker(\partial_1)$ for $N \geq 3$

Limitation of admissible cycles

Admissible cycles do not exhaust $\ker(\partial_1)$. In particular, linear combinations of **non-admissible cycles** may still lie in the kernel.

Karaguler, Mayer Path Homology, 2026

Let I and J be two non-admissible cycles with nonempty intersection.
Then there exists a nonzero weighted element supported on $I \cup J$ that lies in $\ker(\partial_1)$.

Structure of $\ker(\partial_1)$ for $N \geq 3$

Spanning tree viewpoint

Let T be a spanning tree of G_u . Each non-tree edge determines a fundamental cycle. Partition these cycles into:

- ▶ admissible cycles,
- ▶ non-admissible cycles.

Main Theorem

For $N \geq 3$, the kernel of $\partial_1 : \Omega_1^N \rightarrow \Omega_0^N$ is generated by:

- ▶ weighted admissible cycles,
- ▶ weighted elements supported on union of non-admissible cycles with nonempty intersection.

Thank you!